



Tobias Stål

GEOSCIENTIST AND ENGINEER

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Peer-reviewed publications

OVERVIEW Peer-reviewed journal articles in chronological order, including accepted papers and manuscripts currently under review.

SELECTED PUBLICATIONS

2026, Halpin, J. A., N. R. Daczko, J. A. Mulder, A. Maritati, T. Stål — Ediacaran–Cambrian high-strain melt-present deformation of the Archean Charcot Province, East Antarctica. *Tectonics*. doi:10.1029/2026TC009479

2026, Manassero, M. C., K. Selway, A. M. Reading, N. Askey-Doran, J. R. Peacock, F. Ramirez, T. Stål — Magnetotellurics, mantle viscosity, integrated geophysics, glacial isostatic adjustment, mantle water content, East Antarctica. *Journal of Geophysical Research: Solid Earth*. In review.

2026, Manassero, M. C., K. Selway, T. Stål, M. Scheiter, F. S. McCormack, M. Lösing, J. A. Halpin, B. Kulesa, A. M. Reading — Bed topography and subglacial conditions of Denman Glacier, East Antarctica: insights from magnetotelluric data and interdisciplinary studies. *Journal of Geophysical Research: Earth Surface*. doi:10.1029/2026JF009277

2026, Kelly, I. D., A. M. Reading, T. Stål, B. Kulesa, A. García-Jerez, J. Piña-Flores, E. Paolucci, A. Tanzini, R. J. Turner, J. C. Magyar, A. P. Bassom — Determining the character of subglacial sediments in the ice–bedrock interface zone of Antarctica using horizontal-to-vertical spectral ratios (HVSRS) of seismic ambient noise. *Journal of Geophysical Research: Solid Earth*. doi:10.1029/2025JB033029

2026, McCormack, F. S., T. Stål, N. Shao, E. MacKie, A. Fabela Hinojosa, M. Lösing, J. Roberts, S. Ehrenfeucht, C. Dow — Synthetic bed topographies for Antarctica and their utility in ice-sheet modelling. *Philosophical Transactions of the Royal Society A*. doi:10.1098/rsta.2024.0537

2026, Arora, D., N. C. Pant, T. Stål, A. Naik, M. Pandey, R. Gupta — Geothermal heat flow in the Princess Elizabeth Land sector, East Antarctica. *Earth-Science Reviews*. doi:10.1016/j.earscirev.2026.105475

2026, Al-Aghbary, M., M. Sobh, T. Stål, M. O. Awaleh, M. Jalludin, C. Gerhards — Improving geothermal heat-flow predictions and uncertainty quantification using clustering-based quantile regression forests. *Geophysical Journal International*. doi:10.1093/gji/ggag1131

2026, Lösing, M., W. Colgan, T. Stål, J. Ebbing, A. G. Busck, T. Zhang, H. Seroussi, F. McCormack, D. Fahrner, et al. — Community heat-flow recommendations: suitable basal boundary conditions for Greenland and Antarctica in ISMIP7. *GEUS Bulletin*. doi:10.34194/r0w9rf81

2025, Lösing, M., A. R. A. Aitken, J. Ebbing, J. A. Halpin, L. Li, M. Moorkamp, A. M. Reading, T. Stål — Linking tectonics and crustal thermal properties in southwestern Australia and East Antarctica through coupled gravity and magnetic analysis. *Journal of Geophysical Research: Solid Earth*. doi:10.1029/2024JB030770

2025, Abram, N. J., A. Purich, M. H. England, F. S. McCormack, J. M. Strugnell, D. M. Bergstrom, T. R. Vance, T. Stål, B. Wienecke, P. Heil et al. — Emerging evidence of abrupt changes in the Antarctic environment. *Nature*, 644(8077), pp. 621–633. doi:10.1038/s41586-025-09349-5

2024, Stål, T., J. A. Halpin, J. W. Goodge, A. M. Reading — Geology matters for Antarctic geothermal heat. *Geophysical Research Letters*, 51(13), e2024GL110098. doi:10.1029/2024GL110098

2023, Fuchs, S., F. Neumann, B. Norden, G. Beardsmore, P. Chiozzi, W. Colgan, A. P. Dominguez, ..., T. Stål et al. — The Global Heat Flow Database: Update 2023. *GFZ Data Services*. doi:10.5880/fidgeo.2023.023

2022, McCormack, F. S., J. L. Roberts, C. F. Dow, T. Stål, J. A. Halpin, A. M. Reading, M. J. Siegert — Fine-scale geothermal heat flow in Antarctica can increase simulated subglacial melt estimates. *Geophysical Research Letters*, 49(15), e2022GL098539. doi:10.1029/2022GL098539

2022, Stål, T., A. M. Reading, S. Fuchs, J. A. Halpin, M. Lösing, R. J. Turner — Properties and biases of the global heat-flow compilation. *Frontiers in Earth Science*, 10, 963525. doi:10.3389/feart.2022.963525

2022, Reading, A. M., T. Stål, J. A. Halpin, M. Lösing, J. Ebbing, W. Shen, F. S. McCormack, C. S. Siddoway, D. Hasterok — Antarctic geothermal heat flow and its implications for tectonics and ice sheets. *Nature Reviews Earth & Environment*, 3(12), pp. 814–831. doi:10.1038/s43017-022-00348-y

2021, Stål, T., A. M. Reading, J. A. Halpin, J. M. Whittaker — Antarctic geothermal heat-flow model: Aq1. *Geochemistry, Geophysics, Geosystems*, 22(2), e2020GC009428. doi:10.1029/2020GC009428

2021, Sanchez, G., J. A. Halpin, M. Gard, D. Hasterok, T. Stål, T. Raimondo, S. Peters, A. Burton-Johnson — PetroChron Antarctica: a geological database for interdisciplinary use. *Geochemistry, Geophysics, Geosystems*, 22(12), e2021GC010154. doi:10.1029/2021GC010154

2020, Stål, T., A. M. Reading, J. A. Halpin, S. J. Phipps, J. M. Whittaker — The Antarctic crust and upper mantle: a flexible 3D model and software framework for interdisciplinary research. *Frontiers in Earth Science*, 8, 577502. doi:10.3389/feart.2020.577502

2020, Morse, P. E., A. M. Reading, T. Stål — Exploratory volumetric deep Earth visualisation by 2.5D interactive compositing. *IEEE Transactions on Visualization and Computer Graphics*, 28(7), pp. 2641–2653. doi:10.1109/TVCG.2020.3037226

2020, Stål, T., A. M. Reading — A grid for multidimensional and multivariate spatial representation and data processing. *Journal of Open Research Software*, 8(1), pp. 1–7. doi:10.5334/jors.287

2019, Stål, T., A. M. Reading, J. A. Halpin, J. M. Whittaker — A multivariate approach for mapping lithospheric domain boundaries in East Antarctica. *Geophysical Research Letters*, 46(17–18), pp. 10404–10416. doi:10.1029/2019GL083453

2019, Morse, P. E., A. M. Reading, T. Stål — Well-posed geoscientific visualisation through interactive colour mapping. *Frontiers in Earth Science*, 7, 274. doi:10.3389/feart.2019.00274

Reports, outreach, code, datasets

2025, Reading, A. M., T. Stål, I. Kelly — Ice Sheets and Interactions, Dronning Maud Land and Coats Land [Data set]. *International Federation of Digital Seismograph Networks*. doi:10.7914/j9cq-xd60

2025, Stål, T., A. M. Reading, S. Kupis, J. Newlands — Casey–Wilkins Adaptable Array [Data set]. *International Federation of Digital Seismograph Networks*. doi:10.7914/SN/Z9_2021

2024, Stål, T., F. S. McCormack, A. M. Reading, N. Askey-Doran, J. A. Halpin, M. Lösing — Geothermal heat shapes the Antarctic ice sheet from below. *Frontiers for Young Minds*. doi:10.3389/frym.2023.1178537

2023, Fuchs, S., F. Neumann, B. Norden, G. Beardsmore, P. Chiozzi, W. Colgan, A. P. Dominguez, M. R. A. Duque, O. M. Ojeda Espinoza, F. Forster, ..., T. Stål et al. — The Global Heat Flow Database: Update 2023. *GFZ Data Services*. doi:10.5880/fidgeo.2023.023

2022, Stål, T. — Fast calculation of Ripley's K and L functions on a sphere. *Zenodo*. doi:10.5281/zenodo.6865005

Academic positions

ACEAS, University of Tasmania

RESEARCH ASSOCIATE IN COMPUTATIONAL PHYSICS (GEOPHYSICS)

Tasmania, Australia

2022 – Present

University of Tasmania

RESEARCH FELLOW IN ANTARCTIC SEISMOLOGY

Tasmania, Australia

2021 – 2022

University of Tasmania

RESEARCH ASSISTANT

Tasmania, Australia

2020 – 2021

University of Tasmania

CASUAL TUTOR, DEMONSTRATOR, AND RESEARCH ASSISTANT

Tasmania, Australia

2016 – 2021

Education

University of Tasmania

PHD, GEOPHYSICS, GEOLOGY, AND COMPUTATIONAL METHODS

Australia

2021

- Thesis: *The Antarctic lithosphere revealed by multivariate analysis.*
- Supervisors: A. M. Reading, J. A. Halpin, J. M. Whittaker.
- Coursework in statistics and computational geophysics.

University of Copenhagen

MSc, GEOPHYSICS

Denmark

2015

- Thesis: *High-resolution S- and P-wave seismic mapping of near-surface chalk deposits.*
- Supervisor: L. Nielsen.
- Coursework in geophysics, stochastic modelling, and sedimentary basins.

University of Copenhagen

BSc, GEOLOGY

Denmark

2011

- Including studies at UNIS, Svalbard.
- Thesis: *Processes and development of the Faroe North Slide Complex.*
- Supervisors: G. K. Pedersen, T. Nielsen.
- Coursework in geology, palaeontology, geomorphology, Quaternary geology, and Arctic marine geology.

Piteå Music Conservatory (Luleå University of Technology)

AD, SOUND ENGINEERING

Piteå, Sweden

1998

- Coursework in electronics, acoustics, radio broadcasting, music theory, and recording technology.
- Including exchange studies at VGIK, Moscow, Russia.

Unincorporated studies

(162.5 ECTS)

Ongoing

- Linguistics (linguistic typology, sociolinguistics, African languages, Nordic studies, pragmatics, grammar).
- Geology; climate change; non-Nordic geology (the American Cordillera); Arctic petroleum provinces; geohazards; GIS and remote sensing.

Selected conferences, workshops, and hackathons

2026, EGU General Assembly, Vienna, Austria — Aq2 and Kq2.

2025, geoBRIDGE Working Group Hackathon, Perth, Australia — participant and project contributor.

2025, EGU General Assembly, Vienna, Austria — Seismicity of Denman Glacier: constraints on geometry and dynamics.

2024, SCAR Open Science Conference, Pucón, Chile — Aq2: refined geothermal heat-flow map of Antarctica; Exploring Atlas Antarktiki; convened three sessions and co-organised collaborations at Bunger Hills.

2023, EGU General Assembly, Vienna, Austria — Using information entropy to optimise and communicate certainty of continental-scale tectonic models.

2023, Maydena Subglacial Hydrology and Geology Workshop, Maydena, Australia — subglacial heat budget (conceptual investigation); organised and co-chaired.

2021, AGU Fall Meeting, New Orleans and online — modelling Antarctic subglacial geothermal heat transfer in higher resolution; co-convened session in Antarctic geothermal heat.

2021, Australian Earth Sciences Convention (AESC) “Core to Cosmos”, online — data-driven tectonic regionalisation of Antarctica; convened two sessions on data science and machine learning.

2020, SCAR Open Science Conference 2020: Antarctic – Global Connections, online — towards a reconciled heat-flow map for Antarctica (Aq1); Wilkes Land sector and Aurora Basin geology.

2020, AGU Fall Meeting, online — multivariate geothermal heat-flow models of Antarctica, towards Aq2.

2019, Geological Society of Australia SGTSG Meeting, Port Lincoln, Australia — probable geology of the Aurora Basin, East Antarctica.

2019, XIII International Symposium on Antarctic Earth Sciences (ISAES), Incheon, Korea — linking Antarctic geological observations and geophysical data in a probabilistic space.

2018, POLAR2018 Open Science Conference, Davos, Switzerland — multi-domain lithospheric model of East Antarctica.

2018, SCAR TACTical heat-flow workshop, Hobart, Australia — talk and poster on the tectonic structure of Antarctica.

2018, IEEE Antarctic and Southern Ocean Forum (ASOF), Hobart, Australia — framework for processing multivariate Antarctic Solid Earth data.

2018, DataTas seminar, Hobart, Australia — Software construction tools make everything so easy!.

2018, Agile* Subsurface Hackathon, Copenhagen, Denmark — winner of People’s Choice Award.

2017, AGU Fall Meeting, New Orleans, USA — combined constraints on structure and physical properties of East Antarctic lithosphere.

2015, Workshop on 3D reflection seismic processing using open-source software, Rice University, Houston, Texas.

Selected non-academic work experience

University of Tasmania

GRIT FACILITY MANAGER – INSTRUMENTATION

Instrumentation management and Antarctic fieldwork support.

Tasmania, Australia

2021 – Present

Geoneon

GEOSCIENTIST

Marine geoscience, GIS, and data compilation.

Australia

2019 – 2021

COWI

FIELD LEADER AND GEOPHYSICIST

MEP, GPR, and TEM fieldwork, and GIS-based survey design and interpretation.

Denmark

2010 – 2015

Al-Harah Theatre

INVITED TECHNICIAN AND VIDEO DESIGNER

Palestine

2011 – 2012

Norwegian Polar Institute

FIELD SCIENTIST

Supporting biological fieldwork.

Svalbard

2010

Rapid Eye

LIGHT DESIGNER

Including the award-winning performance *QUIPROQUO*.

Denmark

2009 – 2016

STÜ

LIGHTING DESIGNER

Including touring.

Estonia

2009 – 2014

National Theatre of Ghana

TECHNICAL PRODUCER

Ghana

2008 – 2009

The Swedish National Touring Theatre

SOUND/VIDEO/LIGHT DESIGNER AND TECHNICIAN

Sweden

2005 – 2007 (2011)

Circus Cirkör

SOUND ENGINEER

Sweden and world tour

2004 – 2006

The Royal Dramatic Theatre

SOUND DESIGNER

Sweden

2003 – 2004

Swedish Radio (P1, P3)

SOUND ENGINEER

Sweden

1997 – 2004

Uppsala Stadsteater

STAGE MANAGER

Sweden

1998 – 2003

Linds Rör AB

PLUMBING APPRENTICE

Sweden

1993 – 1996

Awards & recognition

2023, Royal Society of Tasmania 2023 Doctoral Award.

2021, Wiley top cited article.

2020, PhD Award for Outstanding Performance during Postgraduate Studies.

2019, Progress in Inclusion, Diversity & Equity.

Selected memberships and service

2021–, SCAR INSTANT Heat Flow Subcommittee

Co-chair

2023–, International Heat Flow Commission (IHFC), IUGG

Member

2020, Australian Antarctic Research Conference (AARC)

*Organising
committee*

2019, Geological Society of Australia SGTSG Meeting

*Organising
committee*

2017–2020, School of Natural Sciences Inclusion, Diversity & Equity Committee

Member

Art collaborations (as a scientist)

2025, T. Stål, A. Reading, K. Verell, DJ Jon Smeathers — Beaker Street: Seismic Dance Party (Hobart), a collaborative science–art performance translating Denman Glacier seismic data into an immersive audio–visual experience.

2025, L. Bleach — *An Infrasonorous Archipelago*, Walk&Talk Biennial “Gestures of Abundance” (Azores), using seismo–volcanic infrasound to create tactile and sonic works.

2025, M. Nieboer — *Through Our Eyes*, Australian Embassy, Paris; contributed cartographic design and geospatial visualisation.

2023, L. Bleach — *soft rock monitoring (the slow seismogenic zone)*, EARTHEN. Better Nature, Cement Fondu (Sydney); transmitting vibrational signals from the SYDS urban seismic monitoring site into gallery architecture.

2021, L. Bleach — *Attenuated ground (the slow seismogenic zone)*, TarraWarra Biennial 2021: Slow Moving Waters; integrating double bass, cast toffee, seismometer, tactile transducers, and slow–earthquake signals.

Other skills and experience

TECHNICAL SKILLS Python (Scikit-learn, OpenCV, TensorFlow, dask, xarray, geopandas), GIS (QGIS, ArcGIS, GDAL/OGR), TeX/LaTeX (TikZ), Bash, Linux, macOS, git, SCons, Madagascar, Seismic Unix, SQL, MAX. Some HTML, C, JavaScript (learning). Experience with Microsoft Office, Adobe Creative Suite, Blender (BlenderGIS), GIMP, Inkscape, Fusion CAD. Working but outdated knowledge of MATLAB, SAC, Emacs Lisp, BASIC.

ONLINE COURSES Drawing Nature, Science and Culture: Natural History Illustration NHI101x (University of Newcastle, 2020); 3D Earth spring school (University of Kiel, 2021); Presenting Data and Information by Edward Tufte (2025).

LANGUAGES English (C2), Swedish (native), Danish (fluent), French (intermediate), Russian (functional), Asante Twi (functional), Norwegian (reading and understanding), Esperanto (inactive), Spanish (learning), Yiddish (learning), Estonian (learning).

CERTIFICATES Coastal Skipper Certificate (SWE); Marine HF/VHF Radio Certificate (SWE); firearms licence and hunting certificate (SWE); PADI Advanced Open Water Diver (including dry suit and Nitrox); Wilderness First Aid (AUS); Australian Approved Arrangements Accreditation Biosecurity (AUS); trip leader and terrain-vehicle driver for the Australian Antarctic Division (e.g. Casey 2023).

PRACTICAL SKILLS Welding, woodwork, diesel engines, sewing, cooking and baking.

OTHER INTERESTS Music, literature and theatre; particular interest in old boats, especially tugs.